

July 1, 2026 11:15 AM GMT

US Economics | North America

Consumption growth has slowed – what gives?

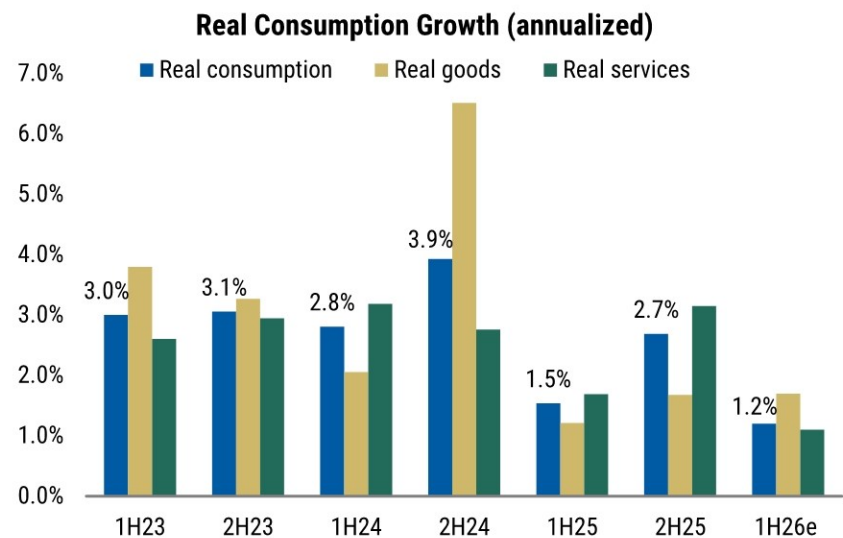
Growth in real consumer spending has slowed year-to-date, led by a slowdown across many services categories. Weak real income growth has weighed on spending. We expect a moderate acceleration in 2H based on our forecast for disinflation ahead.

AlphaWise 

Key Takeaways

- Real consumption growth is tracking a 1.2% annualized pace in 1H vs. last year's 2.1%. Services led the slowdown while growth in durable goods accelerated.
- High prices are weighing on purchasing power. With tariffs, oil, and additional drag from residual seasonality, real income has declined.
- Tax refunds provided a partial offset. We find evidence they could have boosted spending more than expected, explaining the acceleration in durable goods.
- As real income growth recovers, we expect a moderate acceleration in 2H led by services, with consumption for the year ending at 1.7% (4Q/4Q).
- The categories that have accelerated this year are high-income dominated. In 2027, we expect more broadening across spending categories and income cohorts.

Exhibit 1: Real consumption growth has slowed, with a deceleration in services



Source: Census Bureau, Morgan Stanley Research

MORGAN STANLEY & CO. LLC

Heather Berger

Economist

Heather.Berger@morganstanley.com

+1 212 761-2296

Michael T Gapen

Chief US Economist

Michael.Gapen@morganstanley.com

+1 212 761-0571

Arunima Sinha

Global Economist

Arunima.Sinha@morganstanley.com

+1 212 761-4125

Sam D Coffin

Economist

Sam.Coffin@morganstanley.com

+1 212 761-4630

Carolyn L Campbell

Strategist

Carolyn.Campbell@morganstanley.com

+1 212 761-3748

Diego Anzoategui

Economist

Diego.Anzoategui@morganstanley.com

+1 212 761-8573

Lingdi Xu

Economist

Lingdi.Xu@morganstanley.com

+1 212 761-2957

James Egan

Strategist

James.F.Egan@MorganStanley.com

+1 212 761-4715

Raquel Kanner

Strategist

Raquel.Kanner@morganstanley.com

+1 212 761-1103

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

Consumption growth has slowed – what gives?

Growth in real personal consumption has slowed this year, primarily reflecting high inflation that has eroded purchasing power. Residual seasonality in first-quarter inflation data likely exaggerated the slowdown, while tax refunds partially offset the weakness by boosting goods spending more than we expected. As inflation decelerates in the second half of the year, we expect real income growth to turn positive, supporting a moderate recovery in consumption. We expect the acceleration to be led initially by services before broadening somewhat across spending categories and income cohorts next year.

The main downside risk to our view would be if inflation remains elevated, or if tighter monetary policy is needed for inflation to decelerate, both of which would continue to pressure affordability. Upside risks would stem from smaller-than-expected oil effects on goods spending near-term, or a continuation of strong employment growth.

Slower Consumption Growth with a Deceleration in Services

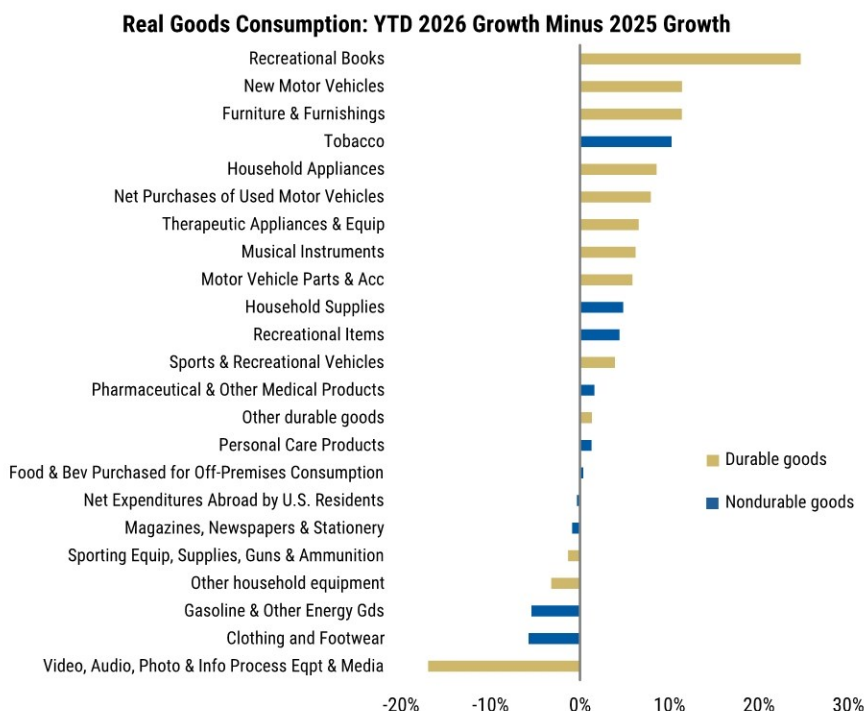
After last week's personal income and spending report and 1Q GDP revisions, we are tracking real consumption growth for the first half of the year at a 1.2% annualized growth rate, a considerable deceleration from the 2.1% growth last year. This includes a very slow 1Q, at 0.5%, followed by an acceleration in 2Q, where we are currently tracking 1.9% q/q saar.

Broadly speaking, the slowdown is in-line with our expectation coming into the year: we expected real consumption growth to slow as higher prices weighed on purchasing power, with larger effects in the first half. This was based primarily on tariff-based pass through effects, though subsequently reinforced by sharply higher energy prices.

That said, the composition of consumption has differed from our expectations. Services consumption growth is running much weaker than its 2025 pace, while goods consumption accelerated; through May, real goods spending was up 1.7% (vs 1.4% last year) while services was up 1.1% (vs 2.4% last year).

The acceleration in goods consumption has been driven by durable goods. Through May real durable goods spending had risen at a 2.3% annual rate compared to just 0.1% 4Q/4Q growth last year. Many of the categories driving the acceleration are big-ticket items, including motor vehicles, furniture, and appliances. Meanwhile, growth in nondurable goods spending, such as clothing and gasoline, has decelerated in real terms versus last year.

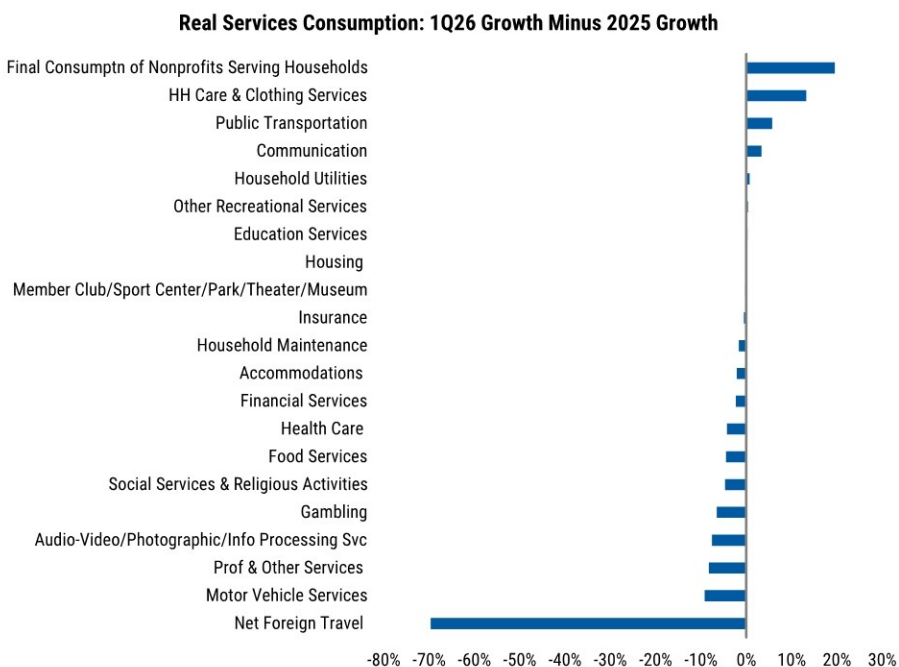
Exhibit 2: Acceleration in real goods consumption growth is largely driven by durable goods



Source: BEA, Haver Analytics, Morgan Stanley Research

In services, most of the April and May data will be revised when the BEA receives the data from the Quarterly Services Survey in August. These revisions are often large, so we focus for now on 1Q data. There, we see that the slowdown in consumption growth was widespread across many services categories, both discretionary and nondiscretionary. These include foreign travel, motor vehicle services, professional services, food services, health care, financial services, and others. The deceleration in health care spending could be related to [the expiration of the ACA credits](#), but we do not think this was the only driver given the broad-based slowdown. As shown in [Exhibit 3](#), more categories beyond health care exhibited weakness.

Exhibit 3: The slowdown in 1Q services consumption growth was widespread across categories



Source: BEA, Haver Analytics, Morgan Stanley Research

Fundamentals Support the Slowing, while Tax Refunds Likely Boosted Goods

We attribute the slowing in consumption growth to two main factors: soft real income growth due to tariff and oil pressures, and residual seasonality in inflation data that exaggerates the softness in both real income and consumption. Tax refunds likely provided a partial offset.

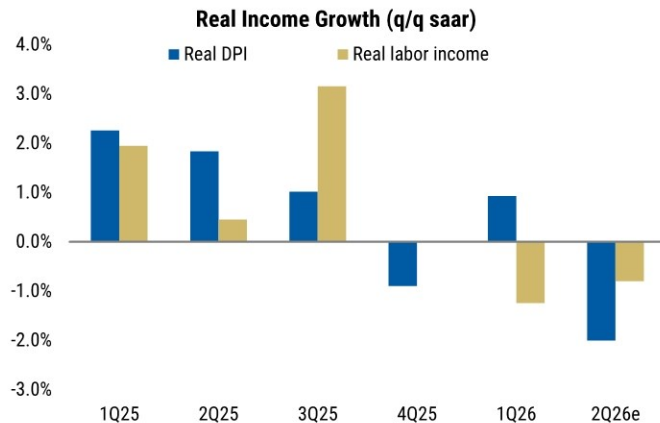
Real income growth is the largest driver of spending. Real disposable income growth was weak in the back half of 2025, down 0.9% q/q saar in 4Q25, and has remained soft this year. For 1H, we are tracking real disposable income down 0.6% (annualized). This is despite the boost from tax changes, so real *labor income* is even softer, tracking -1.0% in 1H. Nominal income growth has slowed from the faster pace of 2023 and 2024, but the weakness over the past couple quarters is largely due to elevated inflation as tariff and oil pressures peak.

Weak income growth alone could justify an even sharper slowdown in consumption. However, consumers have tapped savings to smooth through the price shock. The saving rate, as a percent of nominal personal disposable income, fell from 3.6% at the end of 2025 to 3.0% at the end of May. Wealth gains over the period likely justify part, but not all, of this decline in the saving rate.

We believe residual seasonality has also weighed on measured real income and consumption growth. Since Covid, the seasonal adjustments used by the BEA for inflation have not fully accounted for all seasonal differences. This has resulted in residual seasonality, which "artificially" boosts measured 1Q inflation (with giveback in monthly

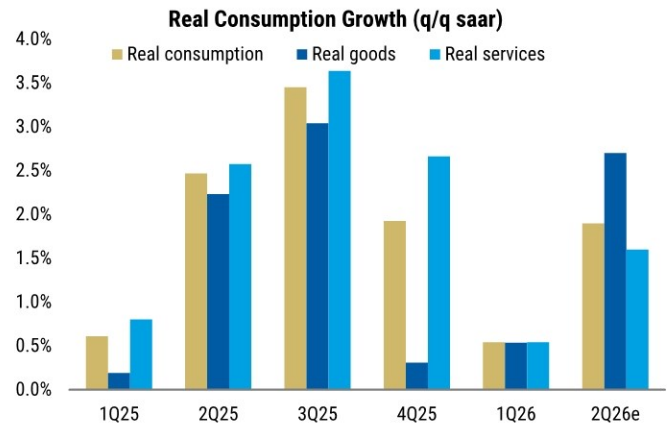
rates of inflation later in the year). Since we look at consumption growth in real terms, this higher inflation mechanically means softer real consumption. We estimate this could have subtracted around 60-80bps from real consumption growth in 1Q. Without it, 1H real consumption growth would likely be around 1.6%; still a deceleration but less extreme.

Exhibit 4: Real income growth has been weak with high inflation



Source: BEA, Haver Analytics, Morgan Stanley Research

Exhibit 5: Consumption growth has slowed but outpaced income growth



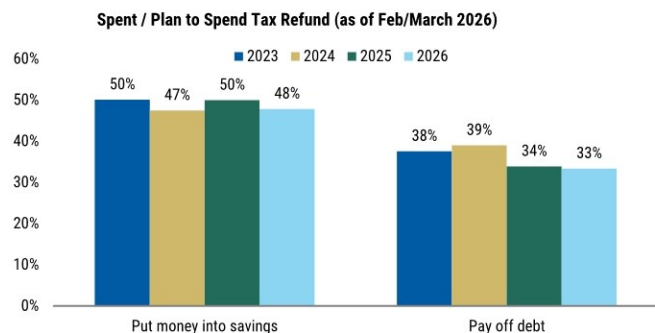
Source: BEA, Haver Analytics, Morgan Stanley Research

We think high tax refunds partially offset the higher prices and help explain the acceleration in goods spending. The acceleration in durable goods spending year-to-date is somewhat surprising given tariffs, oil prices, and higher rates. We had expected tax refunds to boost 1H consumption, but we think there is evidence consumers spent a larger share of refunds more immediately than we expected.

Tax refunds were broadly in-line with [our expectation](#), up 19% YoY (\$57bn). Typically the largest uses of tax refunds are saving and paying down debt. Based on historical patterns, we estimated roughly 35% of tax refunds to be spent within three months, boosting 1Q and 2Q consumption, while the remainder would support spending gradually through the year.

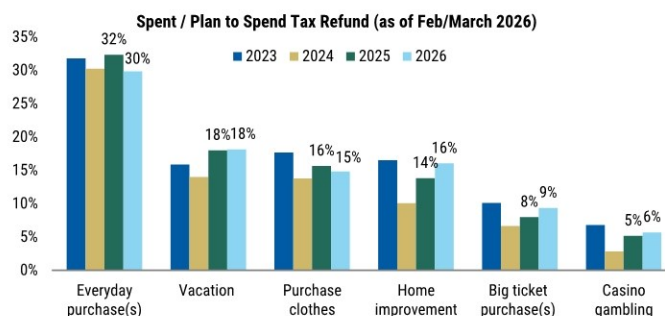
In our monthly AlphaWise Consumer Pulse survey, we ask consumers each February and March how they intend to use their tax refunds. This year, there were slight YoY declines in intentions to save and pay debts, and slight increases in intentions in some spending categories, including home improvement and big ticket purchases. These increases align with the categories of goods spending that have accelerated. Many of these spending areas, such as furniture and motor vehicles, had weak growth in 2H25, so there was likely pent up demand.

Exhibit 6: Intentions to save and pay debts were still high but down slightly YoY



Source: AlphaWise, Morgan Stanley Research

Exhibit 7: While intentions for some spending categories rose YoY

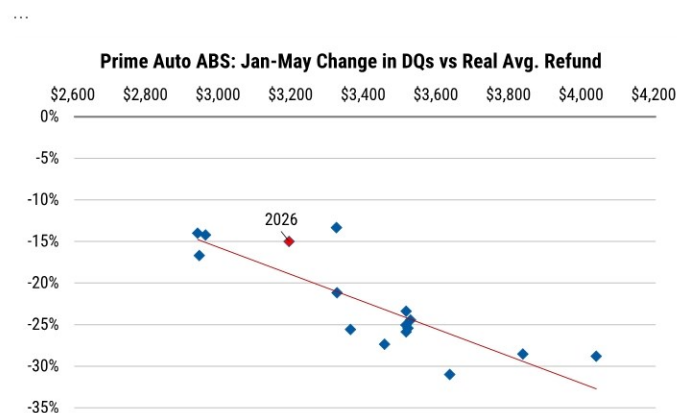


Source: AlphaWise, Morgan Stanley Research

Consumer credit performance also suggests that a larger share of refunds could have been put towards spending instead of towards debt paydown. Historically, larger real tax refunds are associated with larger improvements in auto ABS delinquency rates as consumers use refunds to pay down debt. This year, delinquency improvement in both prime and subprime auto ABS was weaker than the historical relationship would imply. Delinquencies in prime auto ABS declined 15% versus the expected 19%, while delinquencies in subprime auto ABS declined 17% versus expected 21% ([Exhibit 8](#) and [Exhibit 9](#)). Bank credit card performance has improved more noticeably, but those data skew toward higher-income consumers and historically have shown a weaker relationship with refund size.

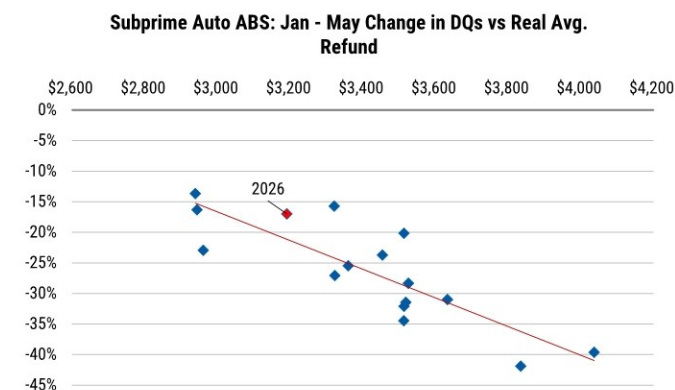
Overall, we think the survey data and ABS data support the idea that a larger share of refunds was spent immediately this year. If roughly 50% of refunds were spent within the first three months, rather than our assumed 35%, and if most of that went towards goods, it would explain the strength in 2Q goods spending versus our forecast.

Exhibit 8: The decline in delinquencies in prime auto ABS from tax refunds was less than the historical relationship would imply



Source: Intex, IRS, Morgan Stanley Research

Exhibit 9: ... and the same was true in subprime auto ABS



Source: Intex, IRS, Morgan Stanley Research

While weak income growth justifies a slowdown in consumption growth overall, the sudden slowdown in services is still surprising to us. Services spending is typically more persistent. While there could have been rotation from goods to services, employment

growth in many services industries has rebounded recently, somewhat contradicting a sudden slowdown in spending. The BEA often revises services spending heavily when they incorporate data from the Quarterly Services Survey. Last year, we saw a similar slowing in 1Q services consumption, followed by a rebound in the subsequent quarters. We expect there could be some upward revisions to April and May services spending when the QSS data is incorporated in August.

Since inflation – like the US economy – is skewed toward services, it may also be the case that residual seasonality affects real services spending more than goods.

We Expect Moderate Re-acceleration Ahead

We continue to expect a slight pickup in consumption growth in 2H, with more strength in services. We expect 3Q and 4Q consumption growth of ~2.0%, around half a percentage point stronger than 1H. Consumption growth for the year moderates to around 1.7% versus last year's 2.1%.

After adjusting for the residual seasonality boost and the tax refund payback, these forecasts are incorporating a slight pickup in underlying consumption growth. We remain more sanguine than the Fed on the outlook for inflation, with headline and core PCE inflation of 3.3% and 3.0% 4Q/4Q this year and falling towards the Fed's 2.0% target thereafter. In contrast, the Fed sees inflation 30-50bp higher this year and less deceleration next year. Based on our outlook for decelerating inflation, we expect real income growth to recover, growing 2.2% (annualized in 2H). This should support spending, but consumption growth remains capped by the lagged effects of high prices earlier in the year.

In the near term, we expect the growth to shift back towards services. If it is true that tax refunds supported goods spending more than expected in 1H, that should mean less support later in the year. Our oil model also suggests the shock should weigh on goods spending with a lag, likely in 3Q. While lower gas prices now may mitigate the effects, we still expect some drag on goods. Payback from residual seasonality may also benefit services.

As we progress into 2027, we expect continued disinflation and productivity gains to support real income and consumption growth, leading to an acceleration back to the 2025 consumption growth pace. At that point, we expect goods spending to recover as well, especially if our expectation for a Fed on hold this year comes to fruition and financial conditions ease.

Structurally, we expect factors leading to greater income and wealth inequality will continue to favor upper-income household spending, but our outlook is for some cyclical improvement and broadening out in consumer spending in 2027 on account of falling inflation.

Before last week's 1Q revisions, risks to consumption and growth looked skewed to the upside, but we now view risks as evenly balanced. Upside risk could stem from a smaller oil shock effect on goods spending than we have factored in, stronger wealth effects, or a continuation of the strong employment gains of the past few months. Upside risk could also materialize if oil prices move lower over the next year on account of a greater re-organization in energy supply lines and energy supply than our commodity

analysts forecast (our commodity analysts foresee dated Brent at \$70-75 between now and the end of 2027).

Key downside risks would be if the currently reported services spending is revised down (and thus the underlying trend is weaker than we have assumed), or if we do not get the deceleration in inflation we expect. If inflation remains elevated, that could also mean tighter monetary policy and financial conditions, which could further weigh on consumption. Finally, inflation may decelerate, but it may take higher interest rates from the Fed to deliver that outcome. Hence, whether its firmer inflation or higher borrowing costs, the affordability crisis and K-shaped consumer remains a key downside risk to our forecast.

Are High-Income Consumers Still Driving Spending?

When the oil shock hit, we pushed out our expectation for a broadening in consumption, given that higher prices would weigh more heavily on low- and middle-income consumers that base their spending decisions primarily on labor market income. We expected the deceleration in consumption growth this year to largely be driven by further slowdown from these cohorts, before some improvement in 2027. Although real-time spending data by cohort are limited, the categories that have accelerated so far this year largely support the idea that high-income consumers are driving consumption growth.

The top 20% income cohort makes up around 40% of all spending. Within that, they make up a similar 40% of services spending, 55% of durable goods spending, and 30% of nondurable goods spending. Among durable goods, the top 20% makes up the largest share of motor vehicle purchases, one of the strongest categories year-to-date. They also make up relatively large shares of other categories that have accelerated this year such as furniture, household equipment, and recreational vehicles.

If tax refunds were used for these purchases, spending could have been more evenly spread across cohorts than usual; typically a smaller share of high-income consumers receive refunds, but for those who do, their refunds are larger. We had expected most of the benefit from the OBBBA provisions to accrue to middle- and high-income consumers, so these groups could have both driven the acceleration in these categories.

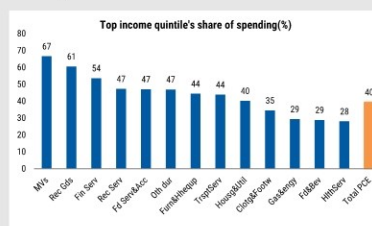
The picture is more mixed for services. Some services categories that are more high-income dominated, such as transportation services, recreation services, and other services, slowed considerably in 1Q. If softer stock prices weighed on high-income spending in 1Q, this could have resulted in the weakness in these categories, and it is hard to know if they have rebounded in 2Q until we get the QSS data.

Overall, the slowing in services may signal a deceleration in spending growth for the high-income group as well. Even if the slowdown proved broad-based, we find it hard to believe the K-shape dynamic has improved this year. Low- and middle-income consumers are disproportionately affected by the decline in real income, and the few spending categories that have accelerated are still high-income

dominated.

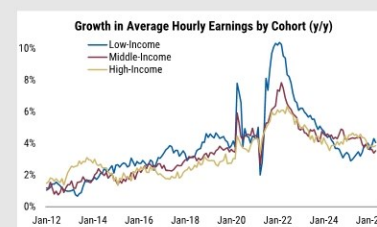
We continue to believe that if our baseline expectation of decelerating inflation and a Fed on hold this year comes to fruition, then there could be recovery ahead for low- and middle-income consumers next year. Lower gas prices and lower inflation more broadly in 2H should lead to a pickup in real income growth, and wage growth for lower cohorts specifically has shown some sign of acceleration recently. However, after several quarters of negative real income growth, we expect the recovery to be gradual, more so driving growth in 2027 as opposed to the next several months. Cuts to Medicaid and SNAP benefits will also be a headwind, offsetting at least part of the rebound for the lowest-income group next year.

Exhibit 10: The top income quintile accounts for a relatively larger share of spending in many durable goods and discretionary services categories



Source: BLS, Haver Analytics, Morgan Stanley Research.
Note: These are using the CEX 2023 shares. The orange bar shows that the top income quintile's share is total PCE is 40%.

Exhibit 11: Nominal wage growth for the lowest income cohort has started to pickup recently



Source: BLS, Haver Analytics, Morgan Stanley Research

Disclosure Section

Mortgage Backed Securities (MBS) and Collateralized Mortgage Obligations (CMO)

Principal is returned on a monthly basis over the life of the security. Principal prepayment can significantly affect the monthly income stream and the maturity of any type of MBS, including standard MBS, CMOs and Lottery Bonds. Yields and average lives are estimated based on prepayment assumptions and are subject to change based on actual prepayment of the mortgages in the underlying pools. The level of predictability of an MBS/CMO's average life, and its market price, depends on the type of MBS/CMO class purchased and interest rate movements. In general, as interest rates fall, prepayment speeds are likely to increase, thus shortening the MBS/CMO's average life and likely causing its market price to rise. Conversely, as interest rates rise, prepayment speeds are likely to decrease, thus lengthening average life and likely causing the MBS/CMO's market price to fall. Some MBS/CMOs may have "original issue discount" (OID). OID occurs if the MBS/CMO's original issue price is below its stated redemption price at maturity, and results in "imputed interest" that must be reported annually for tax purposes, resulting in a tax liability even though interest was not received. Investors are urged to consult their tax advisors for more information. Government agency backing applies only to the face value of the CMO and not to any premium paid.

The information and opinions in Morgan Stanley Research were prepared by Morgan Stanley & Co. LLC, and/or Morgan Stanley C.T.V.M. S.A., and/or Morgan Stanley Mexico, Casa de Bolsa, S.A. de C.V., and/or Morgan Stanley Canada Limited. As used in this disclosure section, "Morgan Stanley" includes Morgan Stanley & Co. LLC, Morgan Stanley C.T.V.M. S.A., Morgan Stanley Mexico, Casa de Bolsa, S.A. de C.V., Morgan Stanley Canada Limited and their affiliates as necessary.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Carolyn L. Campbell; James Egan; Raquel Kanner.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflict/policies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of June 30, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1544	42%	453	49%	29%	757	44%
Equal-weight/Hold	1577	43%	390	42%	25%	769	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	544	15%	89	10%	16%	204	12%
Total	3,668		933			1731	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below. Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures.

For Morgan Stanley specific disclosures, you may refer to <https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch>.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (<http://www.morganstanley.com/terms.html>). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (<http://www.morganstanley.com/terms.html>). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL_Morgan_Stanley_Research_Audit_Report.pdf).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research.

Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V. which is regulated by Comisión Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 14-9169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

The following authors are neither Equity Research Analysts/Strategists nor Fixed Income Research Analysts/Strategists and are not opining on or expressing recommendations on equity or fixed income securities: Michael T Gapen; Lingdi Xu; Heather Berger; Arunima Sinha; Diego Anzoategui; Sam D Coffin.

The following authors are Fixed Income Research Analysts/Strategists and are not opining on or expressing recommendations on equity securities: Carolyn L Campbell; Raquel Kanner; James Egan.

© 2026 Morgan Stanley